In the name of God

HomeWork #2 for "Digital Image processing (DIP)" course

Department of Computer Engineering, University of Guilan,

Asadollah Shahbahrami

Farvardin 18, 1405, Deadline: Ordibehesh 5, 1405, Midnight time

Each question has 10 points

MATLAB/Python-Based Problems

Answer these following questions using both **MATLAB and Python** programming.

1- Write a MATLAB or Python program to read a grayscale image and display its size, data type, and intensity range.

2- Write a program to convert an RGB image to grayscale.

3- Reduce an 8-bit grayscale image to **4-bit quantization (16 levels)**.

4- Write a program to downsample an image by a factor of 2.

5- Upsample an image by a factor of 2 using interpolation.

6- Write a program to generate the negative of a grayscale image.

7- Write a program to extract and print the intensity value of a specific pixel (x,y).

8- Extract the most significant bit (MSB) of an image.

9- Write a program to compute and display the histogram of an image.

10- Write a program to crop a region of size **200×200** from the center of an image.

All answers for each HomeWork should be given:

- In one file, (at the top of file: Student name, student number, HW#2, DIP)
- Format: PDF (typed)
- Name of file: student's name_student number (For example: David_Peter_2020234.pdf)
  Upload in :
  https://quera.org/course/add_to_course/course/27638/ Pass: Image0402